

Panduan Deploy Samasta

Hosting Production di VPS Biznet Gio Cloud + Cloudflare

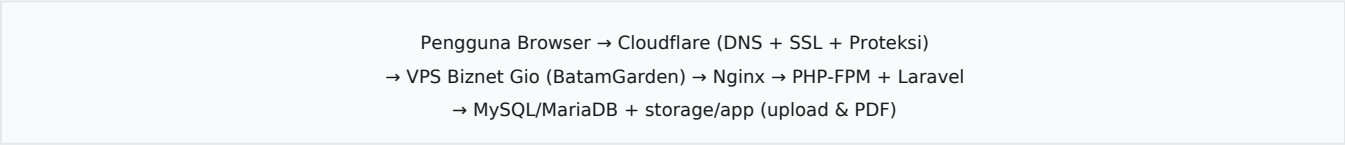
Portal SAMASTA — Disperakimtan Kota Batam

Dokumen	Panduan Deploy VPS Biznet + Cloudflare
Stack Aplikasi	Laravel 12, PHP 8.2+, MySQL/MariaDB, Vite + Tailwind 4, DomPDF
Target Server	Biznet Gio — VPS BatamGarden, Ubuntu 24.04, West Java, 2 vCPU / 4 GB RAM
IP Production	103.150.92.153 (Reverse DNS: ip-153-92-150-103.wjv-1.biznetg.io)
CDN / DNS / SSL	Cloudflare (Free plan cukup untuk awal)
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1. Gambaran Arsitektur



Penting: Cloudflare bukan hosting PHP. Cloudflare berperan sebagai DNS, SSL publik, CDN, dan proteksi DDoS. Aplikasi Laravel tetap harus berjalan di VPS Biznet.

Komponen	Di Cloudflare	Di VPS Biznet
Domain & SSL publik	Ya	Sertifikat origin (Let's Encrypt)
Menjalankan Laravel/PHP	Tidak	Ya
Database	Tidak	MySQL/MariaDB lokal
Upload file & export PDF	Cache static (opsional)	storage/ di server
Cron & antrian job	Tidak	cron + Supervisor

2. Persyaratan Server

Kebutuhan	Spesifikasi Minimum
PHP	8.2 atau 8.3 dengan extension: pdo, mbstring, openssl, tokenizer, xml, ctype, json, fileinfo, curl, zip, bcmath, intl, gd (wajib untuk PDF)
Web server	Nginx
Database	MySQL 8 atau MariaDB 10.11+ (production — jangan pakai SQLite)
RAM	Minimal 2 GB (1 GB terlalu sempit untuk MySQL + Laravel + PDF)
Storage	Minimal 60 GB SSD
Build assets	Node.js 20+ (saat deploy), hasil build disimpan di public/build/
Layanan tambahan	Supervisor (queue worker), cron (scheduler Laravel)

3. Pesan VPS di Biznet Gio

Login ke **biznetgio.com** → Compute → NEO Lite → Create New.

Setting	Rekomendasi
Region	Jakarta — latency terbaik untuk Batam & Indonesia
Operating System	Ubuntu 24.04 LTS
Paket	Minimal 2 vCPU / 2 GB RAM / 60 GB SSD
Snapshot	Centang — backup otomatis sangat disarankan
SSH Key	Import SSH key (lebih aman dari password saja)

Catat setelah VPS aktif: **Public IP**, **SSH username** (bukan selalu ubuntu), dan simpan private key (.pem) dengan aman.

3.1 VPS Production: BatamGarden (aktif)

Item	Nilai
Nama layanan	BatamGarden
OS	Ubuntu 24.04 LTS
Spesifikasi	2 vCPU / 4 GB RAM / 60 GB SSD (MS 4.2)
Region	West Java
Public IP	103.150.92.153
Reverse DNS	ip-153-92-150-103.wjv-1.biznetg.io
SSH Username	GardenBatam
SSH Key Pair	GardenBatam2026 (.pem)
Path aplikasi	/var/www/samasta

Penting: Nama layanan (BatamGarden) dan key pair (GardenBatam2026) **bukan hostname SSH**. Gunakan IP 103.150.92.153 atau reverse DNS Biznet.

4. Akses Pertama & Persiapan Server

4.1 SSH dari Windows (PowerShell)

```
ssh -i "C:\Users\USER\Downloads\GardenBatam2026.pem" GardenBatam@103.150.92.153
```

Alternatif hostname Biznet:

```
ssh -i "C:\Users\USER\Downloads\GardenBatam2026.pem" GardenBatam@ip-153-92-150-103.wjv-1.biznetg.io
```

Jika error permission key di Windows:

```
icacls "C:\Users\USER\Downloads\GardenBatam2026.pem" /inheritance:r  
icacls "C:\Users\USER\Downloads\GardenBatam2026.pem" /grant:r "%USERNAME%:(R)"
```

4.2 Alternatif: Console Biznet

Jika file .pem hilang: Dashboard Biznet → BatamGarden → **Open Console**. Login dengan username **GardenBatam** + password Console Access.

4.3 Persiapan setelah masuk

```
sudo apt update && sudo apt upgrade -y  
sudo timedatectl set-timezone Asia/Jakarta
```

5. Install Stack LEMP (Nginx + PHP + MySQL)

```
sudo apt install -y nginx  
  
sudo apt install -y php8.3-fpm php8.3-cli php8.3-mysql php8.3-mbstring \  
php8.3-xml php8.3-curl php8.3-zip php8.3-gd php8.3-bcmath \  
php8.3-intl php8.3-readline  
  
sudo apt install -y mysql-server  
  
sudo apt install -y git unzip curl supervisor certbot python3-certbot-nginx  
  
curl -sS https://getcomposer.org/installer | php  
sudo mv composer.phar /usr/local/bin/composer  
  
curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash -  
sudo apt install -y nodejs
```

Verifikasi extension GD (wajib untuk export PDF):

```
php -m | grep -i gd
```

Output harus menampilkan: gd

6. Firewall & Keamanan Dasar

Di portal Biznet, pastikan Security Group mengizinkan port **22, 80, 443**. Di server:

```
sudo ufw allow OpenSSH  
sudo ufw allow 'Nginx Full'  
sudo ufw enable  
sudo ufw status
```

Jangan buka port MySQL (3306) ke internet publik. Database hanya boleh diakses dari localhost.

7. Setup Database Production

```
sudo mysql
```

```
CREATE DATABASE samasta CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;  
CREATE USER 'samasta'@'localhost' IDENTIFIED BY 'PasswordKuatMin16Karakter!';  
GRANT ALL PRIVILEGES ON samasta.* TO 'samasta'@'localhost';  
FLUSH PRIVILEGES;  
EXIT;
```

8. Deploy Aplikasi SAMASTA

Setup GitHub: lihat docs/deploy/GITHUB-SETUP.md. Repo: github.com/antonimaulana/samasta

```
sudo mkdir -p /var/www/samasta
sudo chown $USER:www-data /var/www/samasta
cd /var/www/samasta

# Setup deploy key (sekali) – lihat GITHUB-SETUP.md
bash scripts/vps-setup-deploy-key.sh

git clone git@github.com:antonimaulana/samasta.git .

composer install --no-dev --optimize-autoloader

cp .env.example .env
nano .env
```

8.1 Contoh .env Production

```
APP_NAME="Samasta"
APP_ENV=production
APP_DEBUG=false
APP_URL=https://samasta.batam.go.id

APP_LOCALE=id
APP_TIMEZONE=Asia/Jakarta

DB_CONNECTION=mysql
DB_HOST=127.0.0.1
DB_PORT=3306
DB_DATABASE=samasta
DB_USERNAME=samasta
DB_PASSWORD=PasswordKuatMin16Karakter!

SESSION_DRIVER=database
SESSION_ENCRYPT=true
SESSION_SECURE_COOKIE=true

CACHE_STORE=database
QUEUE_CONNECTION=database
FILESYSTEM_DISK=local

LOG_LEVEL=warning
```

Jangan isi ADMIN_PASSWORD di production. Buat akun admin manual via `php artisan tinker` dengan password kuat (minimal 12 karakter).

8.2 Deploy Checklist Aplikasi

```
php artisan key:generate

npm ci
npm run build
rm -f public/hot

php artisan migrate --force
php artisan db:seed --class=WilayahBatamSeeder --force
php artisan db:seed --class=TimPelaksanaSeeder --force
php artisan db:seed --class=DpaDocumentTemplateSeeder --force

php artisan storage:link

sudo chown -R www-data:www-data storage bootstrap/cache
```

```
sudo chmod -R 775 storage bootstrap/cache
```

```
php artisan config:cache  
php artisan route:cache  
php artisan view:cache
```

8.3 Buat Admin Production

```
php artisan tinker
```

```
\App\Models\User::create([  
    'name' => 'Administrator Samasta',  
    'email' => 'admin@sitaman.batam',  
    'password' => bcrypt('PasswordAdminKuat!2026'),  
    'role' => 'admin',  
]);
```

9. Konfigurasi Nginx

Buat file `/etc/nginx/sites-available/samasta`:

```
server {
    listen 80;
    listen [::]:80;
    server_name samasta.batam.go.id;
    root /var/www/samasta/public;
    index index.php;

    client_max_body_size 25M;

    add_header X-Frame-Options "SAMEORIGIN";
    add_header X-Content-Type-Options "nosniff";

    location / {
        try_files $uri $uri/ /index.php?$query_string;
    }

    location ~ \.php$ {
        include snippets/fastcgi-php.conf;
        fastcgi_pass unix:/run/php/php8.3-fpm.sock;
    }

    location ~ /\.(!well-known).* {
        deny all;
    }
}
```

Aktifkan konfigurasi:

```
sudo ln -s /etc/nginx/sites-available/samasta /etc/nginx/sites-enabled/
sudo rm -f /etc/nginx/sites-enabled/default
sudo nginx -t
sudo systemctl reload nginx
```

Contoh lengkap dengan HTTPS tersedia di repo: [docs/Deploy/nginx-sitaman.conf.example](https://github.com/docs/Deploy/nginx-sitaman.conf.example)

10. Hubungkan Domain ke Cloudflare

10.1 Daftar Domain

1. Login ke dash.cloudflare.com → Add a Site
2. Masukkan domain (mis. `batam.go.id` atau subdomain)
3. Pilih plan Free (cukup untuk awal)
4. Ubah nameserver di registrar domain (PANDI) ke nameserver Cloudflare

10.2 Record DNS

Type	Name	Content	Proxy
A	samasta	103.150.92.153	Proxied (orange cloud)

Tunggu propagasi DNS (biasanya 5–30 menit, maksimal 48 jam).

11. SSL & Trust Proxy Laravel

11.1 SSL di Cloudflare

- SSL/TLS → Encryption mode: **Full (strict)**
- Always Use HTTPS: ON
- Minimum TLS Version: 1.2

11.2 SSL di VPS (Let's Encrypt)

```
sudo certbot --nginx -d samasta.batam.go.id
```

APP_URL di .env harus sama persis dengan domain live (contoh: https://samasta.batam.go.id). Ketidakcocokan ini sering menyebabkan CSS/JS tidak termuat.

11.3 Trust Proxy (Laravel di belakang Cloudflare)

Sudah diimplementasi di bootstrap/app.php:

```
$middleware->trustProxies(at: '*');
```

Setelah deploy, pastikan cache config terbaru:

```
php artisan config:clear  
php artisan config:cache
```

12. Cron Scheduler & Queue Worker

Aplikasi SAMASTA memiliki scheduler harian (reminder layanan, digest operasional) dan queue database.

12.1 Cron Scheduler

Jalankan `sudo crontab -e -u www-data` dan tambahkan:

```
* * * * * cd /var/www/samasta && php artisan schedule:run >> /dev/null 2>&1
```

12.2 Queue Worker (Supervisor)

Buat file `/etc/supervisor/conf.d/samasta-worker.conf`:

```
[program:samasta-worker]
process_name=%(program_name)s_%(process_num)02d
command=php /var/www/samasta/artisan queue:work database --sleep=3 --tries=3 --max-time=3600
autostart=true
autorestart=true
stopasgroup=true
killasgroup=true
user=www-data
numprocs=1
redirect_stderr=true
stdout_logfile=/var/www/samasta/storage/logs/worker.log
```

```
sudo supervisorctl reread
sudo supervisorctl update
sudo supervisorctl start samasta-worker:*
```

Contoh lengkap: `docs/deploy/supervisor-sitaman.conf.example`

13. Checklist Go-Live

☐	Item
☐	VPS BatamGarden aktif (103.150.92.153, 4 GB RAM)
☐	PHP 8.3 + extension gd terpasang
☐	MySQL production (bukan SQLite)
☐	npm run build — ada file public/build/manifest.json
☐	Tidak ada file public/hot
☐	APP_DEBUG=false, APP_URL=https://domain-anda
☐	Admin dibuat manual dengan password kuat
☐	php artisan storage:link + permission storage benar
☐	Domain Cloudflare A record → 103.150.92.153 (Proxied)
☐	SSL Let's Encrypt + Cloudflare Full (strict)
☒	trustProxies di bootstrap/app.php (sudah aktif di codebase)
☐	Cron scheduler + queue worker aktif
☐	Test: login admin, halaman publik, upload, export PDF

14. Update Aplikasi ke Depan

```
cd /var/www/samasta
bash scripts/vps-deploy-update.sh
```

Atau manual:

```
cd /var/www/samasta
git pull origin main
composer install --no-dev --optimize-autoloader
npm ci && npm run build
php artisan migrate --force
php artisan config:cache
php artisan route:cache
php artisan view:cache
sudo supervisorctl restart samasta-worker:*
```

15. Estimasi Biaya & Tips Instansi Pemerintah

Komponen	Estimasi Bulanan
Biznet MS 4.2 (2 vCPU / 4 GB)	± Rp 150–250 ribu/bulan (BatamGarden)
Cloudflare Free	Rp 0
Domain .go.id	Sesuai tarif PANDI

Tips Khusus

- Region West Java (BatamGarden) — latency ke Batam ± 30–50 ms
- Aktifkan snapshot Biznet sebelum update besar
- Jangan expose port MySQL ke publik
- File .env jangan di-commit ke Git — simpan backup terpisah
- Pantau log di storage/logs/laravel.log secara berkala
- Upgrade paket NEO Lite jika traffic meningkat

16. Troubleshooting Tampilan Berantakan

Gejala: halaman tampil tanpa CSS/styling. Penyebab umum dan solusi:

Penyebab	Solusi
Belum npm run build	Jalankan npm ci && npm run build di server
Ada file public/hot	Hapus: rm -f public/hot
APP_URL tidak cocok dengan domain	Update .env lalu php artisan config:cache
Cloudflare tanpa trustProxies	Sudah di bootstrap/app.php — jalankan config:cache setelah deploy
Review via trycloudflare.com	URL berubah tiap restart — update APP_URL setiap kali (hanya untuk demo, bukan production)

Perintah Perbaikan Cepat

```
rm -f public/hot
npm ci && npm run build
php artisan config:clear
php artisan view:clear
php artisan config:cache
```

17. Troubleshooting SSH ke VPS

Error / Gejala	Penyebab	Solusi
Could not resolve hostname gardenbatam	Nama layanan Biznet bukan DNS	Pakai IP 103.150.92.153 atau reverse DNS Biznet
Permission denied (publickey)	Key .pem tidak dipakai atau username salah	ssh -i GardenBatam2026.pem GardenBatam@103.150.92.153
Permission denied + username ubuntu	Username Biznet bukan ubuntu	Gunakan username GardenBatam
File .pem tidak ditemukan	Private key hilang	Open Console di dashboard Biznet

UNPROTECTED PRIVATE KEY FILE	Permission key terlalu terbuka (Windows)	Jalankan icacfs — lihat bagian 4.1
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Referensi VPS: docs/deploy/VPS-BATAMGARDEN.md · Nginx: docs/deploy/nginx-sitaman.conf.example · Supervisor: docs/deploy/supervisor-sitaman.conf.example · .env: docs/deploy/.env.production.example